

for farmers. e-Fences—slowly becoming popular in rural India—help them save their crops from animals like elephants.

Vertical farming gaining popularity in smart cities

Vertical farms have the potential to generate more yield than traditional ones in smaller spaces, using trays of crops stacked on top of each other. These also reduce CO₂ emissions and eliminate the need for vehicles to transport food over long distances.

Vertical farms rely on sensor data to establish and maintain ideal growing conditions. Moisture, pH and temperature sensors and more provide thousands of data points that help farmers improve the quality of crops and grow larger quantities.

Food inflation is a persistent problem in Indian economy. While demand patterns are predictable, estimating supply is a challenge. The cause of sudden, unpredictable and sharp rise in food inflation is usually due to lack of requisite and timely supply.

We often see higher price volatility in perishable crops like potatoes, tomatoes, coriander and onions. In recent times, even staples like oil seeds and pulses have shown sharp price fluctuations. The solution to this problem is timely availability of data for sowing, harvesting and producing. IoT-based devices can provide such data accurately.

Use of drones and robots

Smart agriculture can be revolutionised by the use of drones and robotics. According to a recent report, the market for agricultural robots and drones is expected to reach US\$ 35 billion in the next five years.

Researchers are experimenting with different ways of using drones in agriculture. For growers, one exciting application area is crop imaging. Using drones equipped with multispectral sensors, farmers can survey their land, take images that reveal information like fertility of specific patches of soil, determine the amount of water the crops need and more.

In the past, farmers had to rely on satellite imaging to get detailed maps



AgBot II, a solar-powered micro robot (Credit: www.youtube.com)

of their land. This process often took 14 days. With drones, the same can be carried out as and when the farmers want.

Agricultural drones can be used for:

- Monitoring plant health
- Counting plants
- Spraying insecticide and pesticides on crops
- Scheduling seeding/harvesting
- Classifying management zones
- Reducing usage of scarce resources
- Recording data for future analysis
- Increasing yields by using resources effectively

Some real-life applications of robots for smart farming are given below.

Nursery planting. There is a rising need for nursery automation. Companies like HETO Agrotechnics and Harvest Automation provide automation solutions for seeding, potting and warehousing for plants in greenhouses.

Crop seeding. Autonomous precision seeding combines robotics with geo-mapping. A map is generated that shows different properties (quality, density, etc) at every point in the field. A tractor with robotic seeding attachment places the seeds at precise locations and depths, so that each seed has the best chance of growing.

Crop monitoring and analysis. Drone-making companies like Precision Hawk offer farmers packages that

include robotic hardware and analysis software.

Ground-based robots, like Boni Rob, provide detailed monitoring as these can get closer to the crops. Some robots can also be used for tasks like weeding and fertilising.

AgBot II is a solar-powered micro robot. This spraying robot uses computer vision technologies to detect weeds and to spray targeted drops of herbicide onto these.

Lettuce Bot is a thinning robot that uses computer vision to detect lettuce plants as it drives over these. It decides at that moment which plants to keep and which to remove.

Wall-Ye, an autonomous vineyard robot, can prune grapevines. The company behind Wall-Ye has developed a blueberry pruning robot, too.

Researchers are also developing robots for picking, harvesting, shepherding, herding and milking.

Conclusion

Agricultural sensors have been invented to meet the increasing worldwide demand for food. These are simple to use, cheap and help collect and apply data, so that resources can be used most efficiently. The sensors are contributing to solutions that extend beyond farms, including pollution, global warming and conservation.

Autonomous farm vehicles can use improved wireless data transmission and acquisition from smarter, smaller unmanned aerial and unmanned ground vehicles. **EFY**